以下不加证明给出两个定理：

定理1（必要条件）如果$f(P)$在有界几何形体Ω上可积，

则$f(P)$在Ω上有界.

定理2（充分条件）如果$f(P)$在有界几何形体Ω上连续，

则$f(P)$在Ω上可积.

12.2.3 多元函数的积分的性质

以下假设f,g可积，k_1，k_2为常数，

(1) 线性性质

$$\int_{\Omega}[k_1 f(P)+k_2 g(P)]\mathrm{d}\Omega = k_1\int_{\Omega} f(p)\mathrm{d}\Omega + k_2\int_{\Omega} g(P)\mathrm{d}\Omega$$

(2) 分域性质

$$\int_{\Omega_1+\Omega_2} f(P)\mathrm{d}\Omega = \int_{\Omega_1} f(P)\mathrm{d}\Omega + \int_{\Omega_2} f(P)\mathrm{d}\Omega$$

(3) 比较性质

若$f(P)\le g(P)$,则 $\int_{\Omega} f(P)\mathrm{d}\Omega \le \int_{\Omega} g(P)\mathrm{d}\Omega$

也有 $\left|\int_{\Omega} f(P)\mathrm{d}\Omega\right| \le \int_{\Omega} |f(P)|\mathrm{d}\Omega$

从而 $m\Omega \le \int_{\Omega} f(P)\mathrm{d}\Omega \le M\Omega,$

这里M、m为$f(P)$最大最小值.

(4) 中值定理

若 $f(\mathrm{P})$在有界闭区域Ω上连续，则$\exists P_0\in\Omega$,使得

$$\int_{\Omega} f(P)\mathrm{d}\Omega = f(P_0)\Omega$$

(5) 积分与变量的名称无关

$$\iiint\limits_{x^2+y^2+z^2\le 1} f(x,y,z)\mathrm{d}V = \iiint\limits_{u^2+v^2+z^2\le 1} f(u,v,z)\mathrm{d}V$$

$$= \iiint\limits_{y^2+x^2+z^2\le 1} f(y,x,z)\mathrm{d}V$$

重要结论 若几何形体Ω关于$y=x$对称，则

$$\int_{\Omega} f(x,y,z)\mathrm{d}\Omega = \int_{\Omega} f(y,x,z)\mathrm{d}\Omega$$

若几何形体Ω关于$z=y$或者$z=x$对称，也有类似的结论.

例　计算$\iiint\limits_{\Omega}\frac{1+x^2}{3+x^2+y^2+z^2}dV$,其中$\Omega=\{(x,y,z)\big|x^2+y^2+z^2\le 1\}$

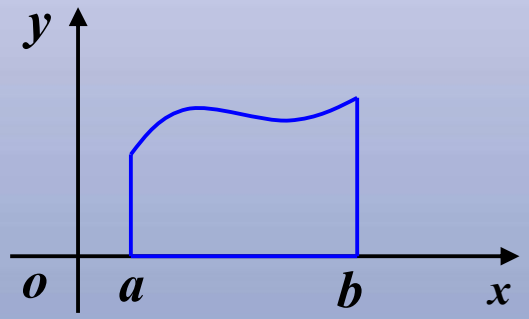
解　由于Ω关于$y=x,z=x,z=y$都对称，则

$$\iiint\limits_{\Omega}\frac{1+x^2}{3+x^2+y^2+z^2}dV=\iiint\limits_{\Omega}\frac{1+y^2}{3+y^2+x^2+z^2}dV=\iiint\limits_{\Omega}\frac{1+z^2}{3+z^2+y^2+x^2}dV$$

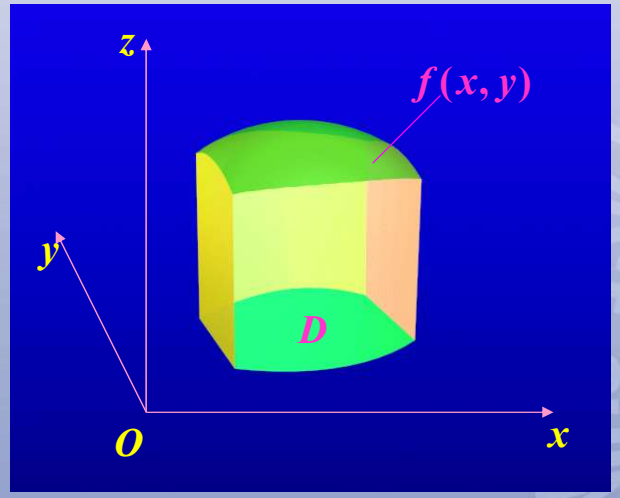
$$=\frac{1}{3}\iiint\limits_{\Omega}\frac{3+x^2+y^2+z^2}{3+x^2+y^2+z^2}dV=\frac{1}{3}\iiint\limits_{\Omega}1dV=\frac{4\pi}{9}$$

12.2.4 多元函数积分的几何意义

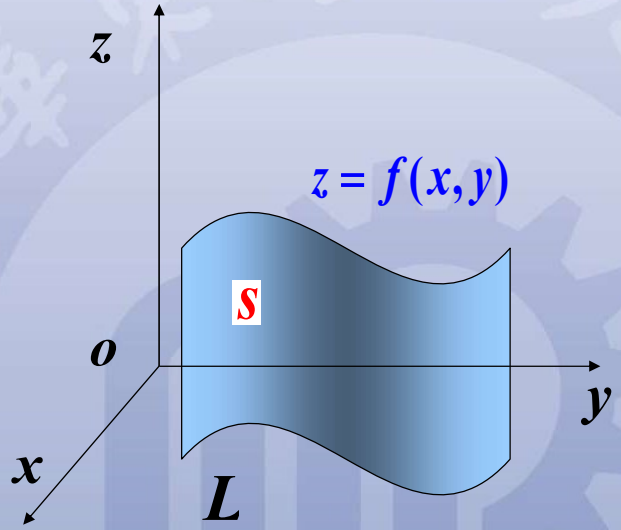
$$\int_a^b f(x)\mathrm{d}x$$



$$\iint_D f(x,y)\mathrm{d}x\mathrm{d}y$$

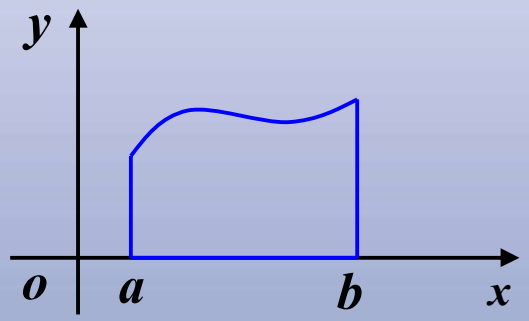


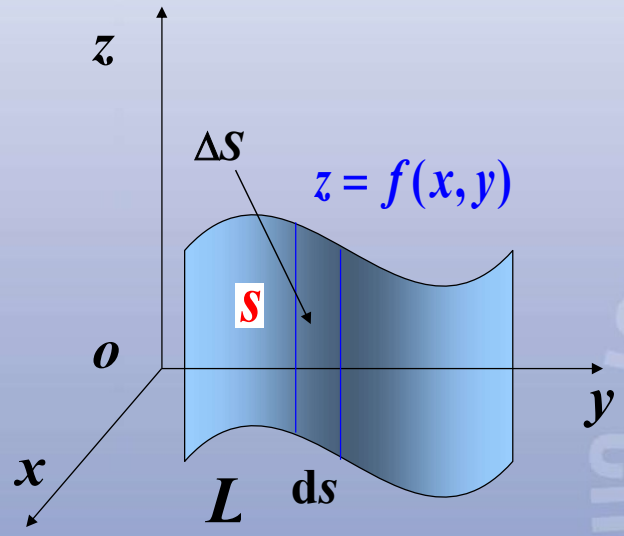
$$\int_L f(x,y)\mathrm{d}s.$$



12.2.4 多元函数积分的几何意义

$$\int_a^b f(x)\mathrm{d}x \qquad \int_L f(x,y)\mathrm{d}s.$$





$$\Delta S = f(x,y)\mathrm{d}s + o(ds)$$

微元法的步骤

1、确定所求量能否通过分割，满足部分之和为一个整体

2、(a)确定对什么区域进行分割（确定积分类型与积分区域）

(b)对应分割所求量（分割可能不唯一）

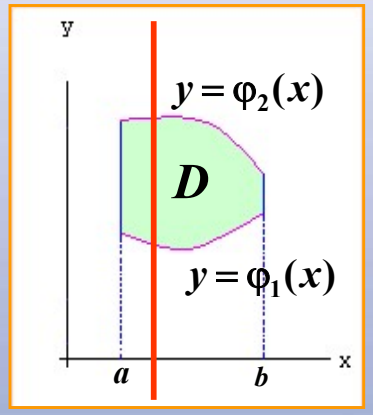
(c)所求量微元用区域微元来表示（所求量微元如果求不出，需要返工）

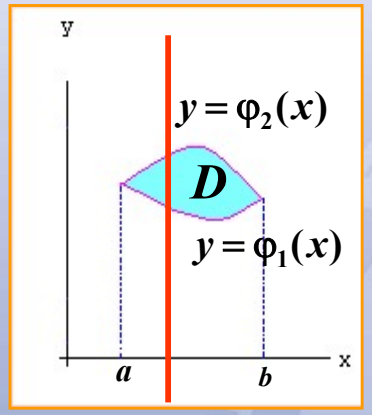
3、积分，求出所求量

一、二重积分的计算

如果积分区域为： $a \le x \le b,\quad \varphi_1(x) \le y \le \varphi_2(x).$

［X－型］



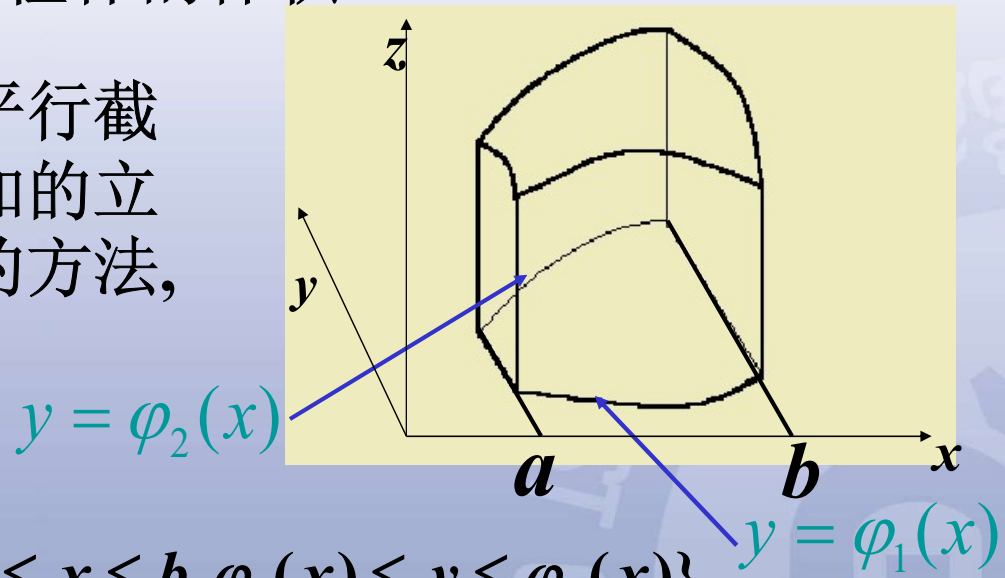


其中函数$\varphi_1(x)$、$\varphi_2(x)$在区间 $[a,b]$上连续.

X型区域的特点： 穿过区域且平行于y轴的直线与区域边界相交不多于两个交点.

$\because \iint\limits_D f(x,y)d\sigma$ 的值等于以 D 为底，以曲面 $z=$ $f(x,y)$ 为曲顶柱体的体积.

应用计算“平行截面面积为已知的立体求体积”的方法,

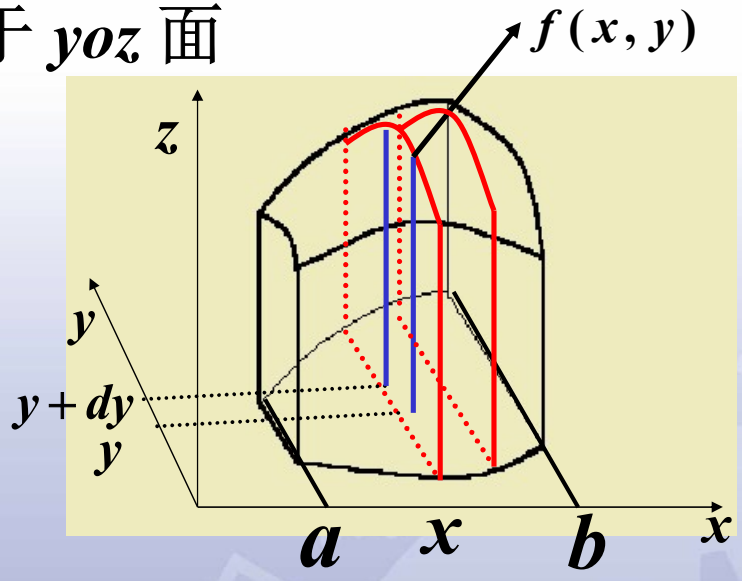


$$D=\{(x,y)\,|\,a\le x\le b,\varphi_1(x)\le y\le\varphi_2(x)\}$$

［X－型］

在D内任取一点x,作平行于 yoz 面的截面.

曲边梯形



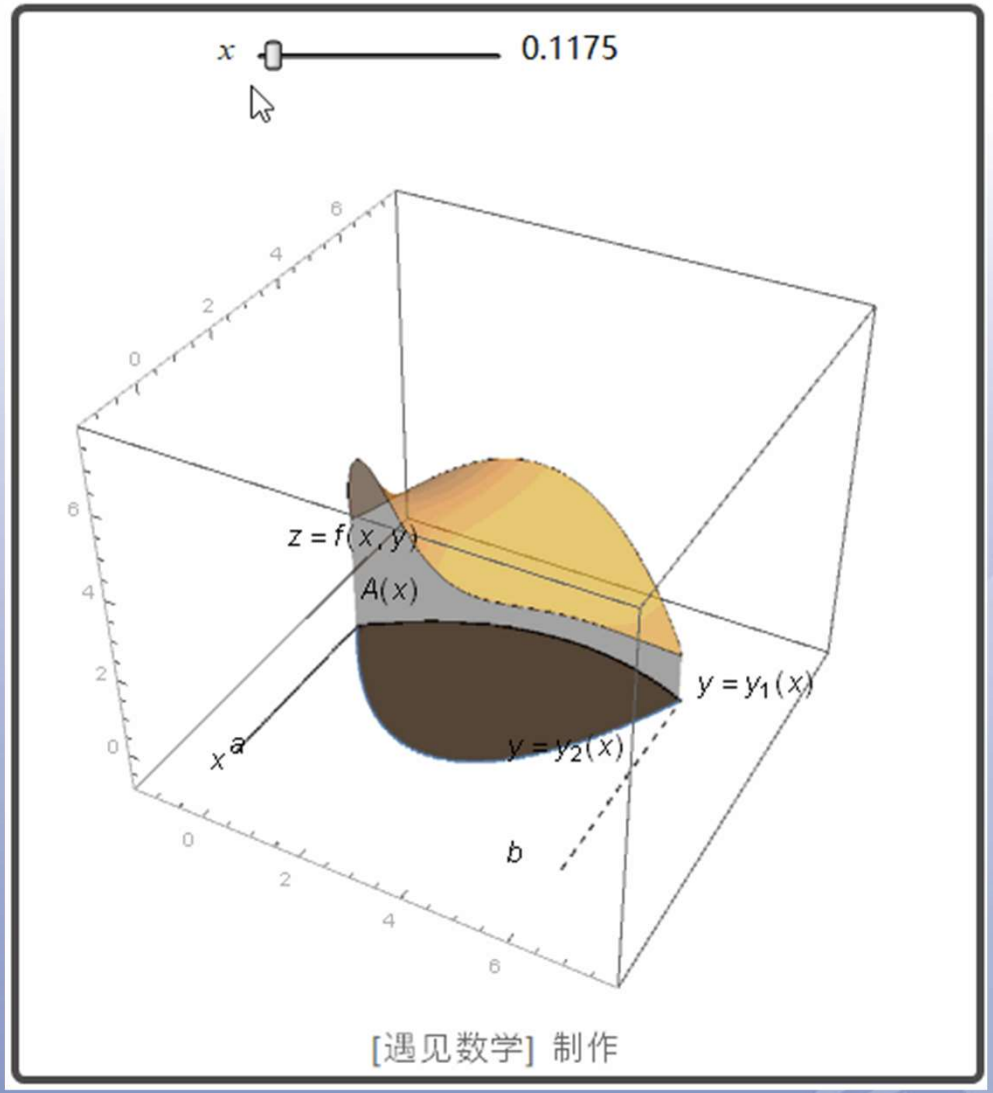
$$A(x)=\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)dy$$

$$V=\int_a^b A(x)dx$$

$$V=\int_a^b [\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)dy]dx$$

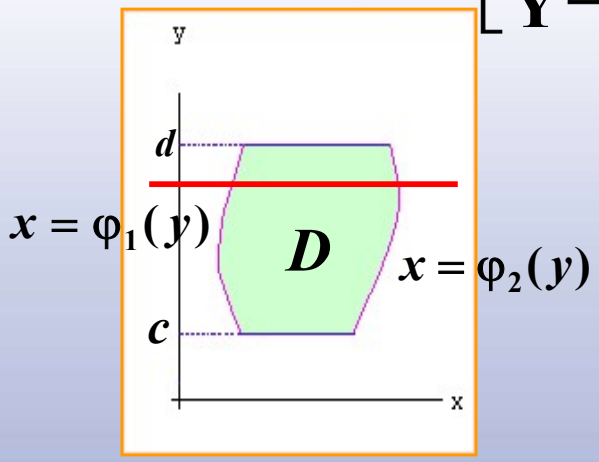
$$\boxed{\iint_D f(x,y)dxdy=\int_a^b dx\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)dy}$$

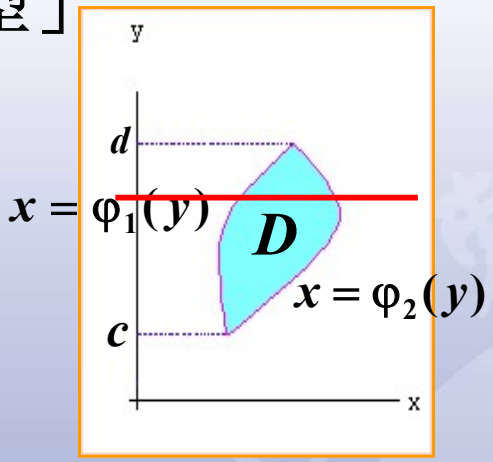
先对y后对x的二次积分


x
0.1175
z = f(x, y)
A(x)
y = y1(x)
y = y2(x)
x
a
b
[遇见数学] 制作

如果积分区域为：$c \le y \le d$，$\varphi_1(y) \le x \le \varphi_2(y)$.

［Y－型］



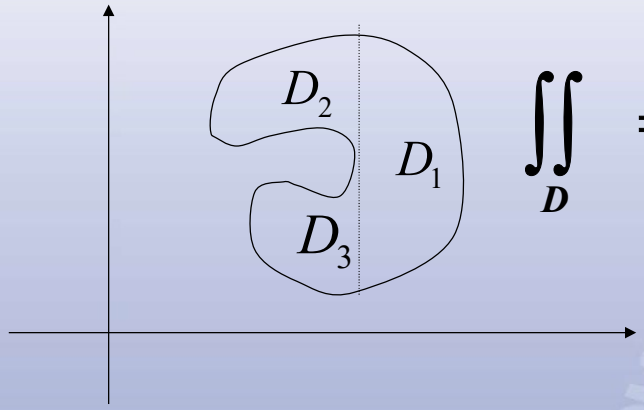


Y型区域的特点： 穿过区域且平行于x轴的直线与区域边界相交不多于两个交点.

$$\iint_D f(x,y)d\sigma = \int_c^d dy \int_{\varphi_1(y)}^{\varphi_2(y)} f(x,y)dx.$$

(2).如果D 既不是X 型域又不是Y 型域,则用平行于坐标轴的直线将D 分成若干子域,利用积分的可加性进行计算.

例如:



$$\iint_D = \iint_{D_1} + \iint_{D_2} + \iint_{D_3}.$$

选择积分域和积分次序是计算的关键

分块越少越好

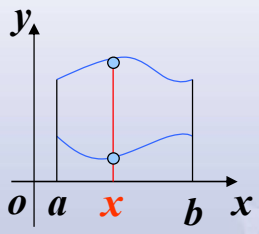
第一次积分要易于计算

一、二重积分的计算

若 $z=f(x,y)$ 在区域 D 上连续，

其中 D：$a\le x\le b$，$g_1(x)\le y\le g_2(x)$，

且 $g_1(x)$，$g_2(x)$ 连续，则有

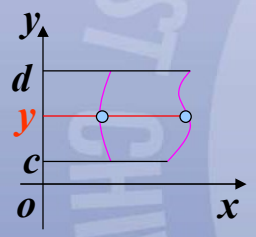


$$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y=\int_a^b \mathrm{d}x\int_{g_1(x)}^{g_2(x)} f(x,y)\cdot\mathrm{d}y$$

若 $z=f(x,y)$ 在区域 D 上连续，

其中 D：$c\le y\le d$，$\varphi_1(y)\le x\le \varphi_2(y)$，

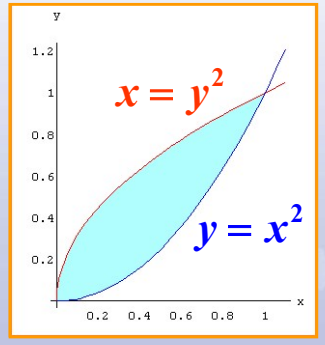
且 $\varphi_1(y)$，$\varphi_2(y)$ 连续，则有



$$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y=\int_c^d \mathrm{d}y\int_{\varphi_1(y)}^{\varphi_2(y)} f(x,y)\cdot\mathrm{d}x$$

例 1 求$\iint\limits_D (x^2+y)dxdy$，其中$D$是由抛物线

$y=x^2$和$x=y^2$所围平面闭区域.



解 两曲线的交点

$$\begin{cases} y=x^2 \\ x=y^2 \end{cases} \Rightarrow (0,0)\ ,(1,1),$$

$$\iint\limits_D (x^2+y)dxdy = \int_0^1 dx\int_{x^2}^{\sqrt{x}}(x^2+y)dy$$

$$=\int_0^1[x^2(\sqrt{x}-x^2)+\frac{1}{2}(x-x^4)]dx=\frac{33}{140}.$$

例 2　计算二重积分 $\iint\limits_D \frac{\mathrm{d}x\,\mathrm{d}y}{(x+y)^2}$，其中

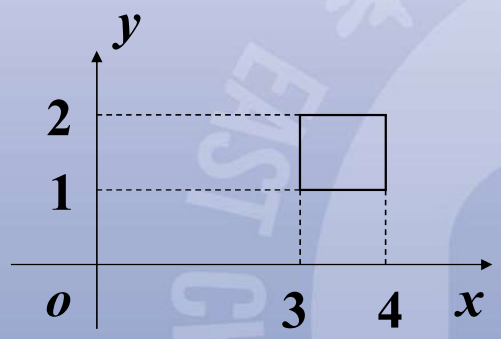
$$D=\{(x,y)\,|\,3\le x\le 4\,,1\le y\le 2\}.$$

解 $\because$ $f(x,y)=\frac{1}{(x+y)^2}$ 在 D 上连续，则有

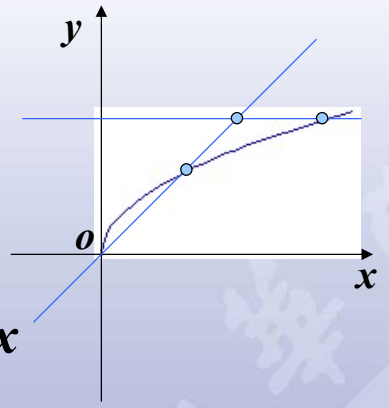
$$\iint\limits_D \frac{\mathrm{d}x\,\mathrm{d}y}{(x+y)^2}=\int_3^4\mathrm{d}x\int_1^2\frac{1}{(x+y)^2}\,\mathrm{d}y$$

$$=\int_3^4\left(-\frac{1}{x+y}\right)\Big|_1^2\mathrm{d}x$$

$$=\int_3^4\left[\frac{1}{x+1}-\frac{1}{x+2}\right]\cdot\mathrm{d}x\ =\ln\frac{25}{24}.$$



例5计算积分 $I=\iint\limits_D \sin\frac{\pi x}{2y}\,\mathrm{d}x\,\mathrm{d}y$，其中积分域 D是由曲线 $y=\sqrt{x}$ 和直线 $y=x$，$y=2$ 所围成.



解 先求得交点 $(1,1)$, $(2,2)$, $(4,2)$.

$$I=\iint\limits_D \sin\frac{\pi x}{2y}\,\mathrm{d}x\,\mathrm{d}y=\int_1^2 \mathrm{d}y\int_y^{y^2}\sin\frac{\pi x}{2y}\,\mathrm{d}x$$

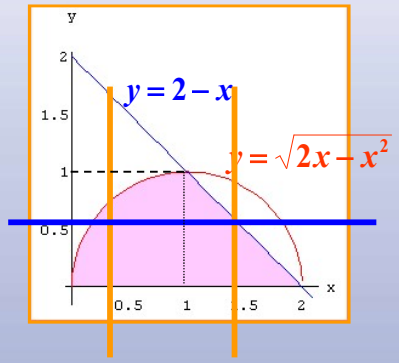
$$=\int_1^2 -\frac{2y}{\pi}\cos\frac{\pi x}{2y}\Big|_y^{y^2}\mathrm{d}y=\int_1^2 -\frac{2y}{\pi}\cos\frac{\pi y}{2}\mathrm{d}y$$

$$=\frac{4}{\pi^3}(2+\pi).$$

例 7 改变积分

$$\int_0^1 dx\int_0^{\sqrt{2x-x^2}} f(x,y)dy+\int_1^2 dx\int_0^{2-x} f(x,y)dy \text{的次序.}$$

解 积分区域如图

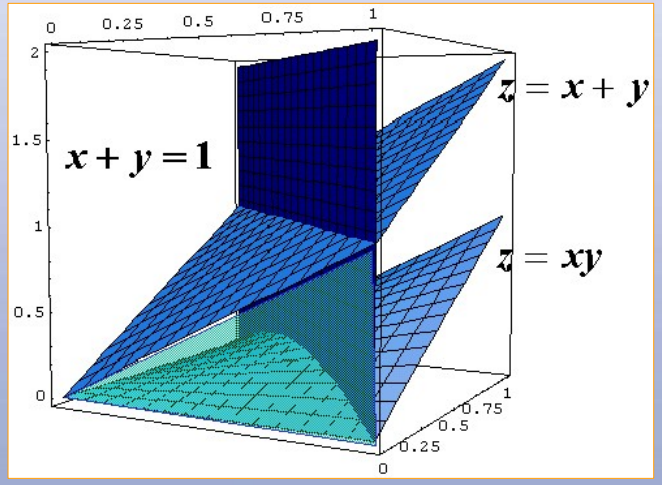


$$\text{原式}=\int_0^1 dy\int_{1-\sqrt{1-y^2}}^{2-y} f(x,y)dx.$$

例 10 求由下列曲面所围成的立体体积，

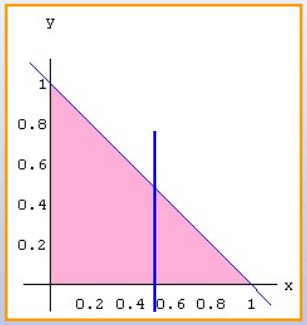
$z = x + y$，$z = xy$，$x + y = 1$，$x = 0$，$y = 0$.

解 曲面围成的立体如图.



该立体能看成两个曲顶柱体相减

所围立体在xoy面上的投影是



$\because\ \ 0 \le x + y \le 1,\ \ \therefore\ \ x + y \ge xy,$

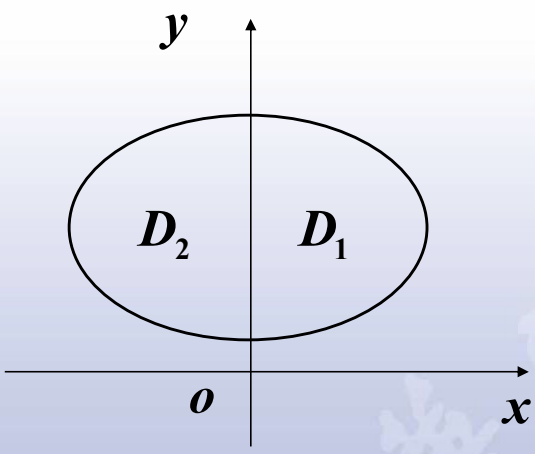
所求体积$V = \iint\limits_D (x + y - xy)d\sigma$

$$= \int_0^1 dx \int_0^{1-x} (x + y - xy)dy$$

$$= \int_0^1 [x(1-x) + \frac{1}{2}(1-x)^3]dx = \frac{7}{24}.$$

二重积分的对称性

1、若D关于y轴对称，



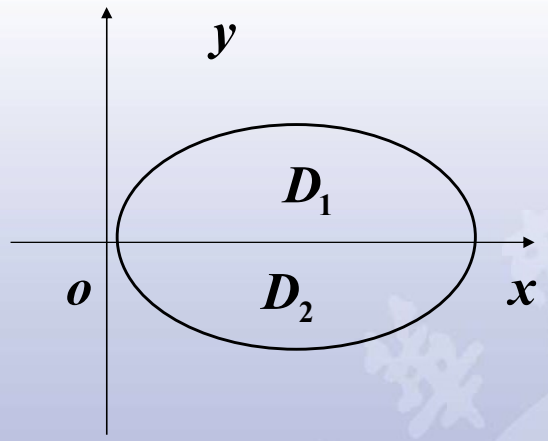
(1)当$f(x,y)$关于x为奇函数，

则 $$\iint_D f(x,y)dxdy = 0$$

(2)当$f(x,y)$关于x为偶函数，

则 $$\iint_D f(x,y)dxdy = 2\iint_{D_1} f(x,y)dxdy$$

2、若D关于x轴对称，



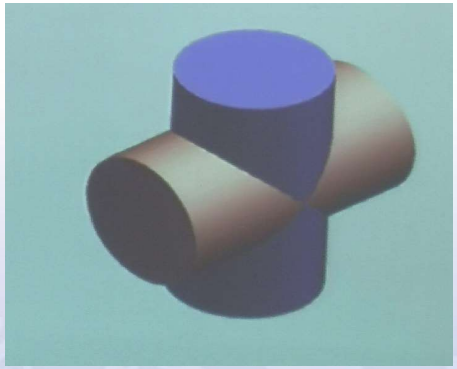
(1)当$f(x,y)$关于y为奇函数，

则 $\iint\limits_D f(x,y)dxdy = 0$

(2)当$f(x,y)$关于y为偶函数，

则 $\iint\limits_D f(x,y)dxdy = 2\iint\limits_{D_1} f(x,y)dxdy$

例12　求$x^2+y^2=R^2$与 $x^2+z^2=R^2$所围成区域的体积



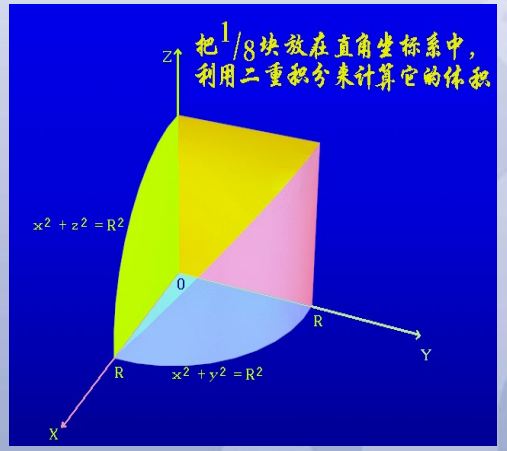
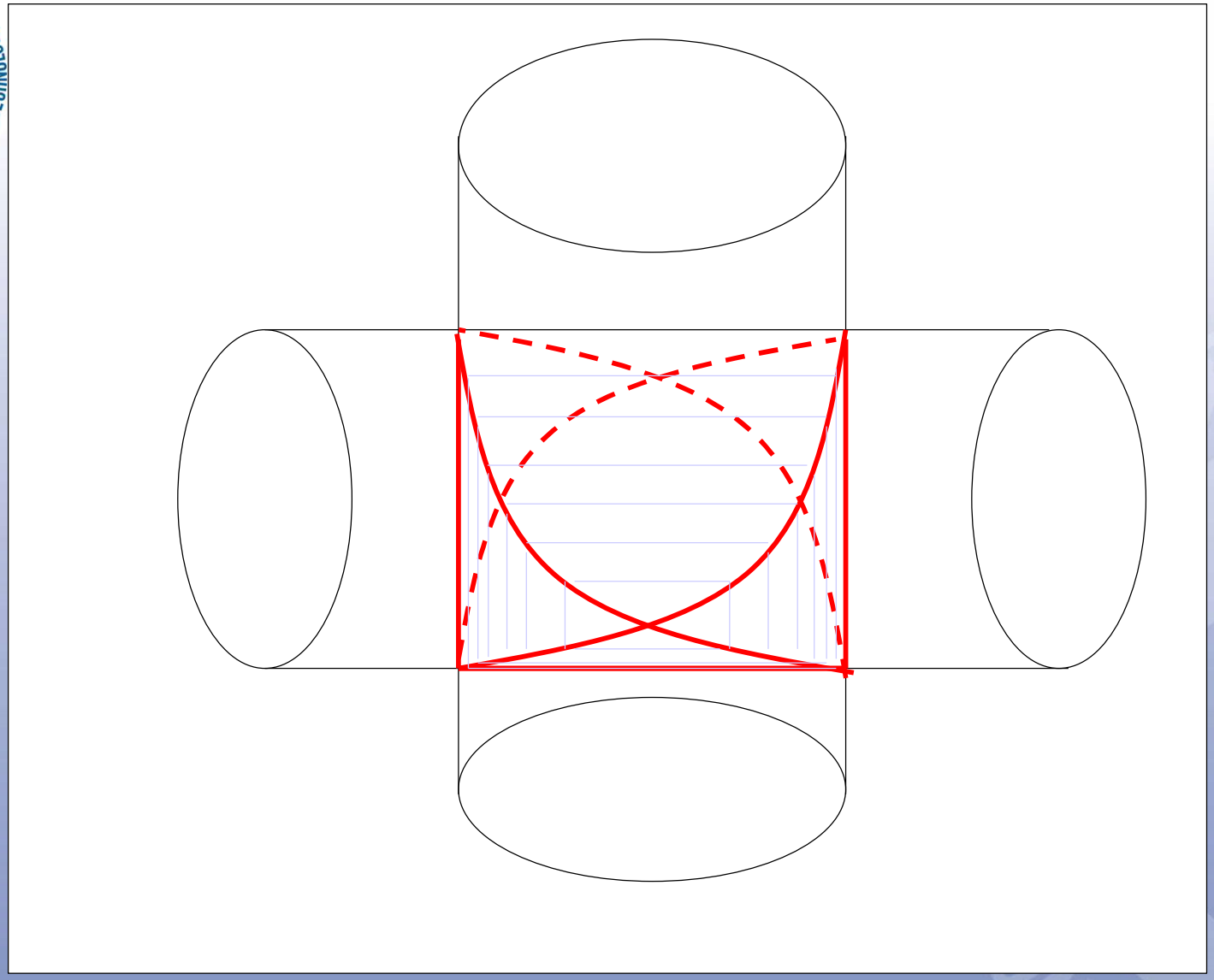
解　则此立体显然关于坐标　平面是对称的.

在第一卦限内的体积为

$$V_1=\iint_D \sqrt{R^2-x^2}\,\mathrm{d}x\,\mathrm{d}y$$

$$=\int_0^R \mathrm{d}x\int_0^{\sqrt{R^2-x^2}}\sqrt{R^2-x^2}\,\mathrm{d}y$$

$$=\int_0^R (R^2-x^2)\,\mathrm{d}x=\frac{2}{3}R^3.$$



故所求体积为 $V=8V_1=\dfrac{16}{3}R^3.$